



SECURE EV CHARGING

**CYBER RISK, GRID STABILITY AND
SOVEREIGN ASSURANCE**

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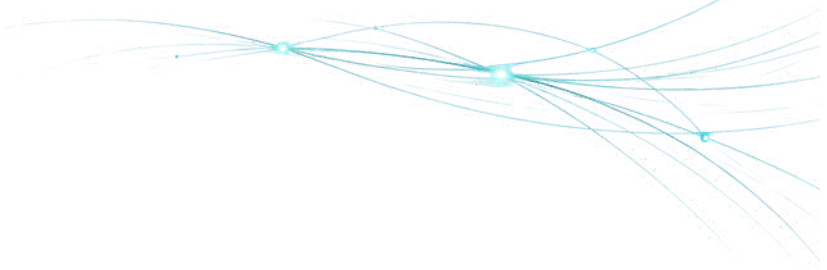
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Introduction

Electric vehicle charging is expanding rapidly across Australia, becoming a familiar part of everyday life. But behind each charger sits a complex, connected system that links vehicles, software platforms and the electricity grid. As these networks grow, they are beginning to influence how electricity is used at scale. This creates a new challenge: if these systems are disrupted or manipulated, the impact may extend beyond individual users to affect the stability of the power system itself.

Electric vehicle charging infrastructure is no longer simply a matter of convenience or consumer choice. It is becoming part of the critical systems that support how Australia generates, distributes and manages electricity. Chargers are connected to cloud services, managed remotely, updated through software, and increasingly coordinated across large networks. This connectivity brings clear benefits, but it also creates new points of exposure.

Today, much of the trust placed in these systems rests with manufacturers and service providers, many of which operate globally. While this model has enabled rapid deployment, it also raises important questions. Who can verify that these systems behave safely under all conditions? Who can intervene if they do not? And how can operators ensure that large numbers of devices act in ways that support, rather than destabilise, the grid?

These questions matter because of scale. A single compromised charger is unlikely to have a significant impact. But thousands of devices, acting in a coordinated or unintended way, can create sudden changes in electricity demand and, with the emergence of Vehicle to Grid (V2G) technology, supply. In a power system that is becoming more distributed and dynamic, such shifts can have real consequences for stability and resilience.

This paper examines the cyber security of electric vehicle charging infrastructure through this broader lens. It explores how risks can propagate from individual devices to networks and, ultimately, to the electricity grid. It also considers the limitations of existing trust models and the growing importance of independent assurance.

Drawing on recent research and practical experimentation, the paper introduces the concept of a Trusted Sovereign Monitor – an approach designed to provide an additional layer of oversight and control for critical energy-connected devices. It aims to ensure that safe operating boundaries can be maintained even when parts of the system are compromised.

The paper concludes by outlining the implications for Australia's energy transition and identifying practical steps that government, industry and standards bodies can take to strengthen the security and resilience of this rapidly growing infrastructure.

Why EV Charging Infrastructure Matters to Grid Security

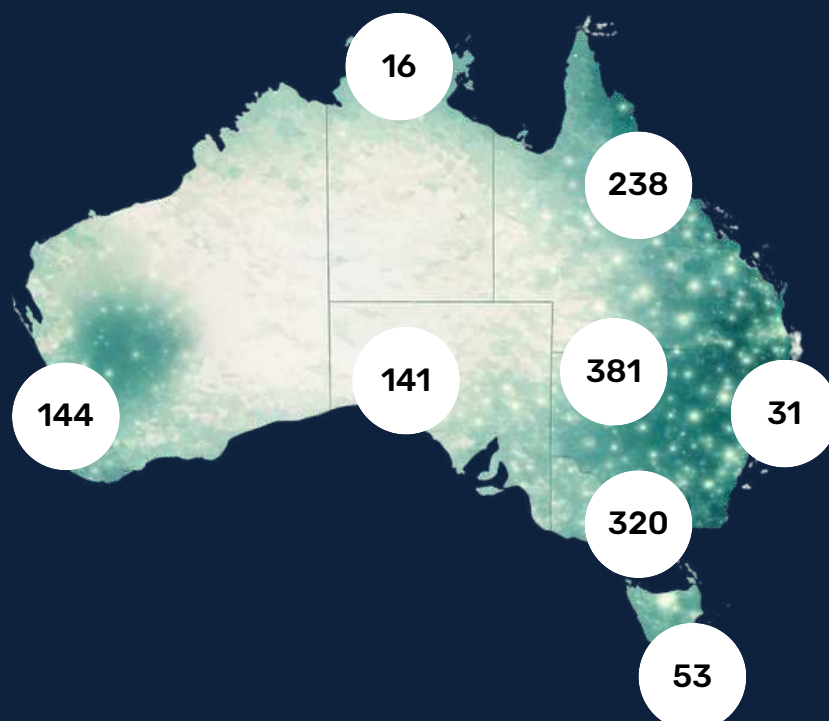
Electric vehicle (EV) charging infrastructure is expanding rapidly across Australia as EV adoption grows and charging networks become more widely available. What was once a supporting service for early adopters is becoming a core part of both the transport system and the electricity system [1].

Today the total installed capacity of fast EV charging infrastructure in Australia represents less than one percent of national generation capacity [2], it is comparable to the output of a mid-sized power station or a credible contingency event in the electricity system. At local and regional levels, coordinated changes in charging behaviour can have material impacts on grid stability.

Chargers, particularly fast chargers, can draw large amounts of power, and their operation is increasingly controlled through software. This means that when charging occurs, how quickly it happens, and how multiple chargers behave can all be influenced centrally or automatically. At small scale, this flexibility is useful. At large scale, it becomes significant. As more chargers are deployed, their combined impact on electricity demand increases.

Large numbers of devices can respond to the same signals at the same time, whether those signals are related to pricing, scheduling, or system conditions. Under normal circumstances, this coordination can help smooth demand and support efficient use of the grid. However, it also introduces the potential for sudden and widespread changes in electricity use.

Fast Public EV Charging Sites as of Q2 2025

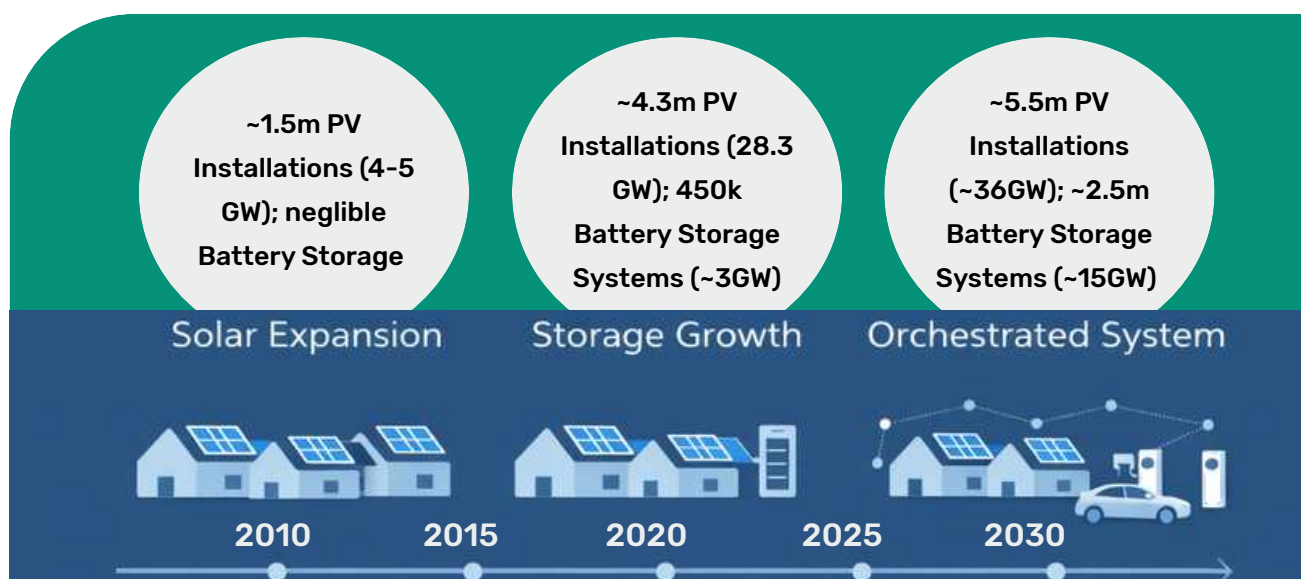




This is particularly important in a power system that is becoming more distributed. Australia already has high levels of rooftop solar and is seeing growing uptake of batteries and other distributed energy resources [3]. In such a system, stability depends on maintaining a careful balance between supply and demand across many smaller, connected assets. Coordinated changes in demand can have noticeable effects.

In this context, EV charging infrastructure should be understood as part of a broader, grid-interactive energy system. Its role is not limited to enabling transport; it is increasingly influencing how electricity flows through the network. As the scale of deployment grows, so too does its relevance to grid operations.

Recognising this role is important for how the infrastructure is designed, managed and secured. It highlights the need to consider EV charging not only in terms of access and convenience, but also in terms of reliability, resilience and system-wide impact.



Based on Rooftop Solar and Storage Report, Clean Energy Council 2026 [3]

Not Just the Charger

At first glance, the cyber security of EV charging may appear to be a question of securing individual devices. In practice, the risk extends well beyond the charger itself. Each charging point operates as part of a broader, interconnected system that includes vehicles, communication networks, back-end platforms, mobile applications and, increasingly, interfaces with energy management systems.

This ecosystem is what enables modern charging networks to function efficiently. Chargers can be monitored remotely, updated through software, and coordinated across large geographic areas. Users can locate, book, access and pay for charging through digital platforms. Operators can manage demand, balance utilisation and respond to system conditions in near real time. These capabilities are central to the value of EV charging infrastructure, but they also introduce multiple points of dependency.

Each connection represents a potential pathway for disruption. Software updates, if not properly governed, can introduce unintended behaviour across many devices simultaneously. Back-end systems that manage large fleets of chargers can become high-value targets, as control over these systems can influence how thousands of devices operate. Communication channels between chargers and management platforms may be exposed to interference or manipulation if not adequately secured.

In addition to technical vulnerabilities, the ecosystem also depends on a complex supply chain. Hardware, firmware and software components are often sourced from multiple providers, many operating internationally. This makes it more difficult to establish consistent assurance across the entire system, particularly over time as components are updated or replaced.

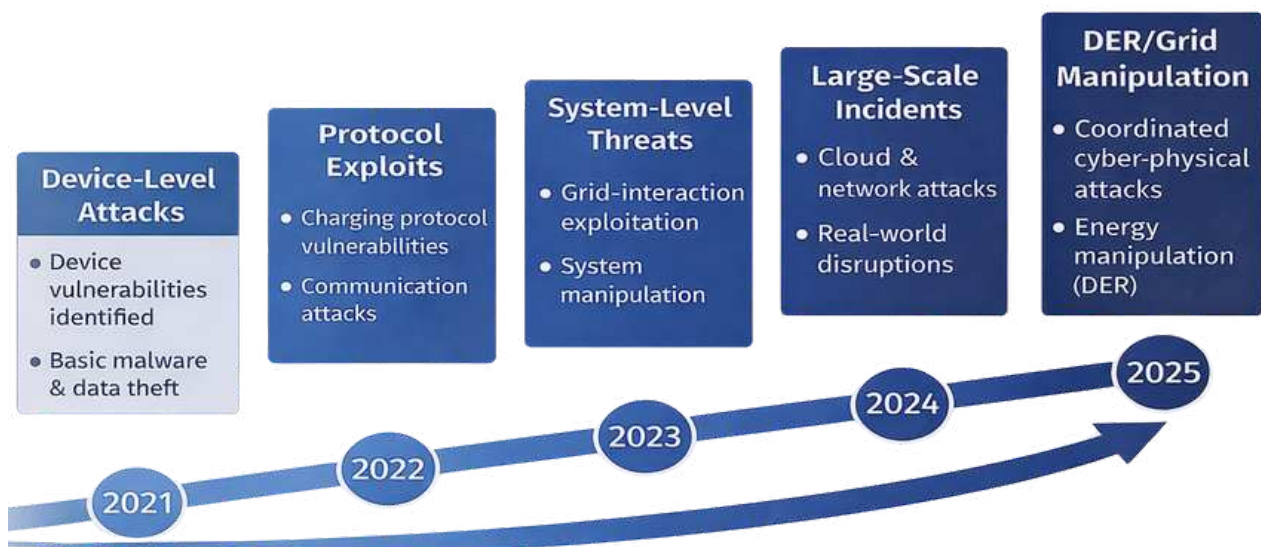
EV charging infrastructure should be understood as a system-level cybersecurity challenge, rather than a collection of isolated devices. Risks are not confined to a single point of failure; they can emerge from interactions between suppliers, users, components and propagate across increasingly complex multi-tenant networks.

Recognising this broader context is important for how security is approached. Measures that focus only on individual devices may not address risks that arise at the system level. Instead, there is a need to consider how trust is established, maintained and verified across the full ecosystem, particularly where coordinated behaviour can have wider impacts.

From Cyber Compromise to Physical Disruption

Cyber events can lead to physical consequences. In many digital systems, a cyber incident may result in data loss, service interruption or financial impact. In contrast, EV charging infrastructure directly controls the flow of electricity. When that control is altered, the effects can extend into the physical operation of the power system.

At the centre of this risk is control over how and when charging occurs. Chargers determine the rate at which electricity is drawn, when sessions start or stop, and how they respond to external signals. These behaviours are governed by software and, in many cases, influenced by remote systems. If that control is disrupted, whether through faulty updates, misconfiguration or malicious interference, the resulting behaviour may no longer align with expected operating conditions.



This creates the potential for rapid and widespread changes in electricity demand. For example, a large number of chargers switching on or off at once, or increasing their charging rate unexpectedly, can produce sudden spikes or drops in load. At the level of a local network, this may place stress on distribution infrastructure. At larger scale, it can contribute to broader system instability, particularly in a grid that is already balancing variable supply and distributed demand.

This progression, from a localised cyber issue to a system-level physical effect, can be understood as a cascade. A disruption at the device level can propagate to a site, from there to a network of chargers, and ultimately to the electricity distribution system itself. The speed at which this can occur is important. Because these systems are digitally connected, changes can be coordinated or replicated almost instantaneously.

In this context, the concern is not only whether individual devices are secure, but whether the system as a whole can tolerate and contain unexpected behaviour. The ability to detect, limit and isolate such effects becomes critical to a future resilient energy grid.

DER infrastructure is now a primary attack surface

The December 2025 cyberattack on Poland's energy system [4] provides one of the first real-world demonstrations of coordinated attacks on distributed energy resources. While the attack did not result in a blackout, it successfully disrupted monitoring and control across multiple renewable energy sites, highlighting how attackers can target the distributed edge of the grid. This incident underscores the growing importance of securing DER infrastructure, where coordinated manipulation of many small assets can have system-level consequences.

Understanding this pathway from cyber compromise to physical disruption is essential for assessing risk in EV charging infrastructure. It highlights why security needs to be considered alongside system behaviour, and why the consequences of failure extend beyond the digital domain into the stability and resilience of the power system.



Australia's Sovereignty Gap

As EV charging infrastructure expands, questions of security are increasingly linked to questions of trust – specifically, who is trusted to design, operate and control these systems over time. Today, much of that trust sits with equipment manufacturers and service providers, many of which operate across global markets and supply chains.

This model has enabled rapid deployment and innovation. It means that critical components of EV charging infrastructure, such as hardware, firmware and back-end management systems, are often developed, maintained and updated outside Australia's direct control. In practice, this creates a reliance on external assurance: operators and regulators must trust that systems behave as intended, both at the point of installation and throughout their operational life.

Recognising this, Australia has begun to establish baseline security expectations for connected devices [5]. Initiatives led by the Department of Home Affairs, in collaboration with the IoT Alliance, are progressing IoT security labelling and certification schemes to improve transparency and uplift minimum security practices across consumer-connected devices [6]. EV charging equipment, particularly residential and small-scale systems, increasingly falls within this category of internet-connected, remotely managed devices.

Relevant Consumer Energy Resources baseline capabilities:

- remote telemetry and monitoring,
- remote updating of device settings,
- response to external control signals,
- trusted communication pathways, and
- the ability to modulate or disconnect in response to grid conditions.

CAPABILITIES

Technical standards and interoperability frameworks are emerging across the energy sector. Protocols such as IEEE 2030.5 (its Australian adoption in AS 5385) [7], CSIP-AUS [8] and related standards define capabilities for interoperability and grid integration.

Many of these requirements are currently voluntary and evolving, with gaps in standards and conformance testing still being addressed. This means that assurance is uneven across devices, vendors and deployments. There are limited mechanisms for independent verification of how devices behave once deployed, particularly at runtime.



Capabilities that enable interoperability and grid integration, such as remote control, software updates, coordinated response, also expand the attack surface and dependency on trusted actors.

For example:

- Remote update capabilities are essential for maintenance and security, but also create a pathway for large-scale change.
- Trusted communication pathways rely on authentication and encryption, but do not by themselves guarantee the integrity of device behaviour.
- Open, Plug & Play device management and communication protocols improve interoperability, but can enable coordinated control across many devices.

Grid operators and regulators may have visibility of aggregate outcomes, such as demand patterns, but not of the underlying logic driving device behaviour or the integrity of software updates applied across fleets. This creates a gap between the minimum security and interoperability compliance, and ongoing, system-level assurance of behaviour.

As EV charging infrastructure becomes more tightly integrated into grid operations, this gap becomes more consequential, and sovereignty is critical. In this context, sovereignty does not imply domestic manufacturing of all components – it refers to the ability to:

- independently verify device behaviour,
- constrain or bound that behaviour within safe limits, and
- intervene where necessary to maintain system stability.

This requires a shift in perspective. Security cannot rely solely on compliance with minimum standards or vendor assurances. Instead, there is a need for operational assurance mechanisms that provide visibility and control at the system level, even when underlying components are heterogeneous, distributed and externally managed.

As EV charging infrastructure continues to scale, closing this trust and sovereignty gap will be essential to ensuring that Australia's energy system remains secure, resilient and under appropriate control.

What this project demonstrated

This project examined EV charging infrastructure through a combination of STRIDE-based threat modelling and cyber-physical co-simulation, focusing on how risks emerge at system level.

The STRIDE assessment [9] identified credible attack pathways across the ecosystem, particularly through firmware and software update mechanisms, backend management systems, and trusted communication channels. The consistent finding is that the most significant risk lies in the ability to introduce common behaviour across large numbers of devices, rather than in isolated compromise.

To explore this, the project developed a co-simulation platform [10] linking EV charging infrastructure with power system dynamics. This enabled the assessment of how coordinated control actions, whether through updates, signals or platform behaviour, translate into physical effects on the grid.

The co-simulation shows that EV charging infrastructure behaves as a coordinated, high-capacity, software-controlled load. When synchronised, this load can produce rapid and material shifts in demand, particularly at distribution and regional levels. These effects emerge without device failure, with systems continuing to operate but in ways that are misaligned with grid conditions.

The analysis demonstrates that impacts scale non-linearly with aggregation and can propagate quickly, leaving limited opportunity for intervention once coordinated behaviour is established. The resulting risk is therefore not simply disruption, but loss of predictable and bounded system behaviour.

Taken together, the project establishes that EV charging infrastructure introduces a system-level, cyber-physical risk driven by coordination and scale, and that current approaches do not provide sufficient visibility or control over how this behaviour manifests in operation.

This provides the basis for introducing mechanisms that can observe, validate and constrain system behaviour at runtime.

A Trusted Sovereign Monitor as a Practical Response

The findings of this project point to a clear requirement: the ability to observe, verify and, where necessary, constrain system behaviour at runtime. This is the role of a Trusted Sovereign Monitor (TSM) [11].

The TSM is not a replacement for existing devices, platforms or standards. It is an additional layer, integrated in future devices, providing independent oversight of behaviour as EV charging infrastructure interacts with the grid. Its purpose is to ensure that coordinated, software-driven actions remain within safe and predictable operating boundaries, even where underlying components are heterogeneous and externally managed.

The TSM addresses the gap between capability and assurance. While current frameworks enable devices to communicate, respond to signals and operate in a coordinated manner, they do not provide a mechanism to independently verify that these actions remain aligned with system requirements. The TSM provides this capability through three functions:

Visibility:

- The TSM provides a system-level view of how distributed assets are behaving in aggregate, drawing on telemetry and operational data. This is not limited to observing individual devices, but understanding emerging patterns of coordinated behaviour across networks.

Validation:

- Observed behaviour is assessed against expected operating envelopes, such as limits on aggregate demand changes, response rates or grid conditions. This allows the system to identify when behaviour is unexpected, unsafe or inconsistent with defined policies, regardless of whether the underlying cause is error or malicious activity.

Intervention:

- The TSM can act to constrain or modulate behaviour, ensuring that system-level impacts remain within acceptable bounds. This does not require full control of individual devices, but the ability to influence aggregate behaviour, for example by limiting response rates or dampening coordinated actions.

From a sovereignty perspective, the TSM provides a mechanism for independent assurance within a globally connected system. It allows Australia to retain the benefits of international supply chains and innovation, while ensuring that the behaviour of critical infrastructure remains observable, controllable and aligned with national requirements. The TSM represents a shift in approach. Rather than attempting to eliminate all sources of risk, it accepts that systems will be complex, connected and partially externalised.

Policy Implications

Policy settings are moving in the right direction. National initiatives such as the National Electric Vehicle Strategy [12] and the National Consumer Energy Resources (CER) Roadmap [13] are designed to accelerate EV and CER uptake, expand charging infrastructure, and improve interoperability across distributed devices. State strategies in New South Wales [14], Victoria [15], South Australia [16] and Western Australia [17] are doing the same, treating EV charging as an enabling layer of the energy transition rather than simply a transport service.

This matters because EV charging infrastructure is becoming more deeply integrated into the electricity system. DCCEE's recently released initial 18 CER device requirements [5] make that direction explicit. They include capabilities such as remote telemetry, remote updating of device settings, response to external signals, trusted communication pathways and fail-safe operation, and they explicitly include EV supply equipment within scope.

These reforms are necessary. They support interoperability, consumer choice and more flexible grid operation. But they also normalise the capabilities that make coordinated behaviour possible at scale. Remote updating, fleet-level communications and dynamic response are not peripheral functions; they are becoming baseline expectations for connected energy devices. The result is that integration capability is increasing faster than runtime assurance capability.

This is where cyber policy and national security begin to converge. Through the Cyber Security Act [18] and the Security Standards for Smart Devices Rules 2025 [19], Australia has introduced mandatory cyber security standards for most consumer smart devices acquired in Australia, with the Rules in force since 4th March 2026 after a 12-month transition. In parallel, Home Affairs and IoT Alliance Australia have advanced the Security Labelling Scheme for Smart Devices [6], a voluntary scheme intended to improve transparency and uplift baseline security across connected products.

For EV charging equipment, that baseline matters. In residential and small commercial settings, chargers are increasingly internet-connected, remotely managed and software-dependent devices. But the policy issue does not end with consumer-device security. Once these assets become grid-interactive CER, they begin to affect critical infrastructure. At that point, the question is no longer only whether a charger is secure enough for a home network. It is whether thousands of such devices, acting through shared protocols, platforms and update paths, remain observable, verifiable and governable once integrated into the energy system.



National security implications are unavoidable. Much of the EV charging ecosystem relies on international OEMs, foreign cloud services, offshore software development and globally distributed supply chains. Australia can continue to benefit from those supply chains, but it also inherits their dependencies. Where critical infrastructure relies on hardware, firmware and backend systems that are developed, signed, updated or operated outside Australia's direct control, the exposure is not only technical – it is strategic. In a high-CER electricity system, these dependencies create potential pathways for foreign interference, coercion, compromise or disruption affecting critical infrastructure.

This does not require assuming hostile intent from every vendor or jurisdiction. The issue is structural. If trust is concentrated externally, then assurance is also externalised. That may be acceptable for ordinary consumer devices. It is much harder to accept for assets that are increasingly able to influence feeder behaviour, aggregate demand and local system stability. In that context, foreign dependence is not simply a procurement issue; it becomes a question of critical infrastructure governance and sovereign control.

International regulation is moving in the same direction. The European Union's Cyber Resilience Act [20] is designed to impose lifecycle cyber obligations on products with digital elements, including requirements around secure design, security updates and vulnerability handling. Its significance lies in shifting expectations from one-off compliance at point of sale to ongoing accountability across the product lifecycle. That logic is highly relevant to EV charging and CER, where software-defined behaviour persists long after deployment.

The implication for Australia is that existing frameworks are building an important foundation, but they do not yet close the gap between device compliance and system assurance. Standards, labelling and certification can improve minimum security and interoperability. They do not by themselves provide independent assurance over how fleets of devices behave once deployed, updated and coordinated in real-time.

That is the policy gap this project exposes. As EV charging infrastructure and CER become more connected, more responsive and more deeply embedded in the grid, policy must move beyond minimum standards alone. It must also address how Australia can maintain independent visibility, validation and intervention capability over globally sourced, software-defined infrastructure with material impact on Australia's power systems.

Conclusions

Electric vehicle charging infrastructure is becoming a foundational component of Australia's energy system. As deployment accelerates, it must be understood as a connected, software-defined, and grid-interactive system with system-level implications.

The key risk is not the compromise of individual devices, but the potential for coordinated behaviour to emerge at scale. The combination of remote control, shared platforms and standardised communication pathways creates the conditions under which large numbers of devices can respond in a synchronised manner. At sufficient scale, this behaviour can produce material and rapid changes in electricity demand, impacting grid stability.

These risks arise directly from the capabilities required to enable a flexible and interoperable energy system. Remote updating, dynamic response and coordinated control are essential features of modern Consumer Energy Resources, including EV charging infrastructure. They also introduce shared dependencies and common control pathways, often extending across global supply chains and external service providers.

Current policy and standards frameworks are establishing an important foundation. Interoperability requirements, baseline security measures and emerging certification schemes are improving consistency and trust at the point of deployment. However, they remain primarily focused on design-time assurance. They do not yet provide sufficient capability to observe, verify and govern behaviour in operation, particularly where coordination occurs across large numbers of devices.

This creates a gap between what the system can do and what can be assured in practice. Addressing this gap requires a complementary layer of system-level assurance that operates alongside existing standards and controls. The concept of a Trusted Sovereign Monitor provides a practical pathway. It enables independent visibility of system behaviour, validation against defined operating boundaries, and the ability to constrain coordinated actions where necessary.



For this approach to be effective at scale, it must be supported through standards and policy frameworks. This includes:

- defining reference architectures and interfaces for Trusted Sovereign Monitoring,
- encouraging OEMs and platform providers to design systems that can be externally observed and constrained,
- ensuring interoperability between monitoring layers and existing CER standards, and
- enabling authorised national or jurisdictional entities, for example system operators, aggregators or designated agencies, to provide independent assurance of system behaviour.

This is not about centralising control or limiting innovation. It is about ensuring that, within a globally connected and rapidly evolving ecosystem, Australia retains the ability to independently verify and govern the behaviour of infrastructure that has system-level impact.

Australia's energy system is entering a phase where cyber security, system stability and national sovereignty are increasingly intertwined. EV charging infrastructure, and more broadly Consumer Energy Resources, sit at the intersection of these domains. Ensuring that they remain secure and reliable in operation will be essential to the resilience Australia's broader energy system in an increasingly uncertain world.

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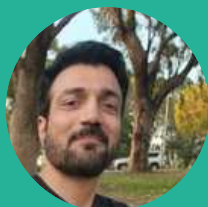
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